

# TAL KLEIN

052-6338989 | kleintal7@gmail.com | [LinkedIn](#) | [GitHub](#)

## SUMMARY

M.Sc. researcher in Software Engineering (Meitar Excellence Program, GPA 92), specializing in NLP, LLMs, and causal inference. First-author paper in submission to *Artificial Intelligence and Law* (Springer). Hands-on experience designing LLM-based extraction pipelines on 4,000+ Hebrew legal documents, with strong foundations in classical ML, deep learning, and statistical modeling.

## EDUCATION

**Ben-Gurion University of the Negev** 2024 – Jul. 2026 (Expected)  
*M.Sc. Software & Information Systems Engineering — Meitar Excellence Program* GPA: 92

- Thesis:** LLM-driven feature extraction and causal inference for sentencing analysis in Hebrew court verdicts

**Ben-Gurion University of the Negev** 2021 – 2025  
*B.Sc. Data Engineering* GPA: 86

- Key courses:** Machine Learning, Deep Learning, NLP, Computer Vision, Algorithms, Statistics, Big Data

## EXPERIENCE

**Data Scientist & ML Researcher — NLP & Causal Inference** 2024 – Present  
*Socially Embedded Lab, Ben-Gurion University* Beer Sheva, Israel

- Built an end-to-end NLP pipeline extracting 30+ structured features from 4,000+ unstructured Hebrew court verdicts using GPT-5-mini batch inference; achieved **0.92% F1** against a manually-annotated test set of 125 documents
- Designed and iteratively refined extraction prompts through error analysis on a held-out dev set, improving extraction accuracy by **11 percentage points** over a zero-shot baseline
- Conducted causal inference using propensity score matching (R, MatchIt) with sensitivity analysis (Rosenbaum bounds, E-values) and MICE imputation to quantify sentencing disparities

## PUBLICATIONS

Klein, T., et al. "Examining Bias in Sentencing Decisions with LLM-Powered Control over the Facts." *Under review, Artificial Intelligence and Law (Springer), 2026.*

## SELECTED PROJECTS

**Wikipedia-Scale Information Retrieval Engine** | *Python, BM25, TF-IDF, PageRank, Word2Vec, GCP*

- Implemented and deployed a hybrid search engine over **6.4M Wikipedia documents** on GCP, combining BM25, TF-IDF, PageRank, and PageViews into a weighted final ranking
- Optimized retrieval through multi-threading, heap-based candidate ranking, and impact ordering — reducing average query latency **from 9.8s to 1.24s (8x speedup)** while maintaining **Precision@10 of 0.63**

**The Sound of Silence — LLMs vs. Human silence Annotators Research** | *Python, LLMs, Hebrew NLP, AssemblyAI*

- Built an end-to-end Hebrew-dialogue NLP pipeline (audio diarization via AssemblyAI, dynamic silence detection, custom web annotation platform with audio-playback tracking) and released the first annotated dataset of conversational silences (315 segments, 1,425 labels) on Hugging Face
- Benchmarked Gemini-2.0-flash on 6-class silence classification: model achieved **Cohen's k=0.13** with audio vs. **k=0.26** for untrained human annotators, demonstrating a measurable gap in LLM understanding of non-verbal pragmatic cues

**Multimodal Next-Frame Video Prediction** | *Python, PyTorch, BLIP-2, YOLO*

- Designed a multimodal model fusing YOLO object-detection embeddings with BLIP-2 captioning to predict narrative continuity across video frames

## TECHNICAL SKILLS

**Languages & Tools:** Python (advanced), R, SQL, Bash, Git, Docker, GCP, PostgreSQL  
**ML & DL:** PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, TensorFlow  
**LLMs & Agents:** LLM APIs, prompt engineering, structured outputs, RAG, LangGraph, LLM evaluation  
**NLP:** Hebrew NLP, Transformers, Hugging Face, embeddings  
**Computer Vision:** OpenCV, YOLO, BLIP-2, CLIP  
**Data & Stats:** Pandas, NumPy, Selenium, ETL, MICE imputation, propensity score matching, Bayesian methods, A/B testing  
**Spoken:** Hebrew (native), English (fluent)

## ADDITIONAL EXPERIENCE

**IT Support & Help Desk** 2020 – 2021  
*Matrix* Israel

- Monitored fault patterns and improved incident response across organizational systems

**Communications & Operations NCO** 2017 – 2020  
*Israel Defense Forces (IDF)* Israel

- Led system upgrades and supervised a team of 8 soldiers managing data and network operations

## VOLUNTEER

**Volunteer Puppy Raiser (Foster Family Program)** 2025 – Present  
*Israel Guide Dog Center for the Blind*